the heating or not – and ‘knowing’ that I exist? How can the continuous exchange of information in nature produce *us*, and our thoughts?

The problem is wide open, with numerous fine solutions currently under discussion. This, I believe, is one of the most interesting frontiers of science, where major progress is about to be made. Today new tools allow us to observe the activity of the brain in action, and to map its highly intricate networks with impressive precision. As recently as 2014 the news was announced that the first complete (mesoscopic) detailed mapping of the brain structure of a mammal had been achieved. Specific ideas on how the mathematical form of the structures can correspond to the subjective experience of consciousness are currently being discussed, not only by philosophers but also by neuroscientists.

An intriguing one, for instance, is the mathematical theory being developed by Giulio Tononi – an Italian scientist working in the United States. It’s called ‘integrated information theory’, and is an attempt to characterize quantitatively the structure that a system must have in order to be conscious: a way, for example, of describing what actually changes on the physical plane between when we are awake (conscious) and when we are asleep but not dreaming (unconscious). It’s still at the developmental phase. We still have no convincing and established solution to the problem of how our consciousness is formed. But it seems to me that the fog is beginning to clear.

There is one issue in particular regarding ourselves which often leaves us perplexed: what does it mean, our being free to make decisions, if our behaviour does nothing but follow the predetermined laws of nature? Is there not perhaps a contradiction between our feeling of freedom and the rigour, as we now understand it, with which things operate in the world? Is there perhaps something in us which escapes the regularity of nature, and allows us to twist and deviate from it through the power of our freedom to think?

Well, no, there is nothing about us that can escape the norms of nature. If something in us could infringe the laws of nature we would have discovered it by now. There is nothing in us in violation of the natural behaviour of things. The whole of modern science – from physics to chemistry, and from biology to neuroscience – does nothing but confirm this observation.

The solution to the confusion lies elsewhere. When we say that we are free, and it’s true that we can be, this means that how we behave is determined by what happens within us, within the brain, and not by external factors. To be